

Youssef EL OTMANI

AI & Machine Learning student (M2) · Université Paris-Saclay

Looking for a 6-month internship in AI & machine learning, starting March 2027

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youyou354.github.io
Interactive portfolio, with projects you can run

EDUCATION

MSc in Artificial Intelligence 2025 — 2027
Université Paris-Saclay — Faculty of Sciences — taught entirely in English
Year 2 in progress. Year 1: machine learning, deep learning, optimisation, NLP, information retrieval, logic & constraints, a research project and an internship.

Double BSc — Mathematics and Computer Science 2022 — 2025
Institut Galilée, Université Sorbonne Paris Nord — Villetaneuse, France
Mention Très Bien in year one, Bien in years two and three. Class representative, robotics club.

Erasmus exchange semester January — June 2025
Mälardalen University — Västerås, Sweden
Courses in artificial intelligence, computer algebra and computer graphics.

EXPERIENCE

Intern, AI team May — August 2026
Fédération Agirc-Arrco
Four months helping build the federation's AI platform and the tools for specific use cases, and supporting the people who use them.

Association secretary Academic year 2025/2026
Association Irisé
Preparing, organising and hosting events; managing the mail, the social networks and the administrative paperwork.

Hospitality officer June — September 2025
Centre d'Affaires Central Seine

Private tutor Academic year 2024/2025
Maths, physics-chemistry, biology and computer science

SKILLS

Programming languages
Python · C · Java · OCaml · JavaScript · HTML · CSS · XML · SageMath · OPL · Rocq

Machine learning
PyTorch · Hugging Face Transformers · Keras · scikit-learn · XGBoost · NumPy · pandas · NLP · Deep learning · SMOTE · Codabench

Operating systems
Linux · Windows

Tools
Git · Docker · Azure · Langfuse · OpenTelemetry · Word · Excel · PowerPoint

LANGUAGES

Arabic native
French DALF C1
English IELTS C1
Swedish A2
Korean A1

CERTIFICATION

Google IT Support Professional Certificate

INTERESTS

Archery · Swimming · Volleyball · Robotics

SELECTED PROJECTS

Personalised learning with a Hierarchical Reasoning Model
M1 research project · team of 4
A Hierarchical Reasoning Model trained on ASSISTments (198 skills) predicts a student's next skill with 69.5 % accuracy, against about 36 % for the best prompted LLM.
Python · HRM · LLM prompting · Gradio · Supabase

Pollinator identification: a Codabench challenge
M1 · team of 6 · 19/20
A Codabench competition built from 528 INRAE videos: macro-F1 scoring, unseen camera angles in phase 2. I wrote the ingestion and scoring programs and the starting kit.
Codabench · Python · scikit-learn · EfficientNet features

Detecting AI inpainting: AEROBLADE on TGIF
M2 · team of 5
Ran the training-free detector AEROBLADE zero-shot on 4,459 TGIF images (Stable Diffusion 2, Photoshop), thresholds calibrated by cross-fitting: ROC-AUC 0.505, a clear negative result.
Python · PyTorch · diffusers · LPIPS · pandas

Named entities in French medical text
M2 · pair
17 taggers on the QUAERO corpora: LSTMs and CNNs on six word embeddings against fine-tuned transformers. CamemBERT-bio comes first, F1 68.4.
Python · PyTorch · Hugging Face Transformers · gensim · CamemBERT

Email thread summarization with event triggers
M1 · team of 4
Fine-tuned DistilBART on Enron threads with silver summaries distilled from GPT-4.1 nano and Gemini, with and without event triggers; ROUGE and BERTScore.
Python · Hugging Face Transformers · DistilBART · ROUGE · BERTScore

Does attention help DRAW draw?
M1 · team of 3
An ablation of DRAW's spatial attention on MNIST, EMNIST and Omniglot: an attention grid too small for the data blurs thin strokes; enlarging it closes the gap.
Python · Deep learning · LSTM · VAE

University timetabling with constraints
M1 · pair
A university timetable in OPL / CP Optimizer, from feasibility to three weighted criteria (gaps, heavy days, days on campus), checked on 45 test instances.
OPL · IBM CP Optimizer · Constraint programming · LaTeX

Artificial intelligence labs — Mälardalen
Erasmus
Search (BFS, DFS, A*), a genetic algorithm and ant colonies for the travelling salesman, minimax for Mancala, and an MNIST network at 93.4 %.
Python · Keras